Number Codes (Set B)

Year 4 · **Mark Scheme**

Lesson Worksheet — Answers

Purple — Accessible

1. KPT

2. 754

3. B) OCK

4. GVO

5. 648

6. D) 248

7. AZB

8. 465

9. A) 184, C) 102 and E) 136 are correct (each starts with the digit 1, which codes to T).

10. A) GBE

11. ECU

12. 861

13. C) 421

14. WAR

15. 123

Green — Standard

16. 23 (9 × 2 = 18, 18 + 5 = 23)

17. 8 ((22 + 2) ÷ 3 = 8)

18. C) 13

19. 20 ((6 + 4) × 2 = 20)

20. 10 ((12 − 7) × 2 = 10)

21. A) 7

22. 27 (rule: ×4 then −1; 7 × 4 − 1 = 27)

23. 7 (rule: ×2 then +6; (20 − 6) ÷ 2 = 7)

24. C) 7, D) 8 and E) 9 are correct (codes 22, 25 and 28)

25. E) 38

26. 21 ((10 − 3) × 3 = 21)

27. 6 ((31 − 1) ÷ 5 = 6)

28. B) 28

29. 32 (rule: ×3 then +2; 10 × 3 + 2 = 32)

30. 5 (rule: +1 then ×2; (12 ÷ 2) − 1 = 5)

Red — Challenge

31. 475 (1→4, 4→7, 2→5)

32. 376 (reverse the add-3 rule: 6−3=3, 0−3→7, 9−3=6)

33. A) 729 (2→7, 7→2, 4→9)

34. 385 (583 reversed)

35. 614 (416 reversed)

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36. A) 729 (927 reversed)

37. 952 (5→9, 1→5, 8→12→2)

38. 729 (reverse the add-4 rule: 1→7, 6→2, 3→9)

39. B) 249 and D) 461 are correct — a digit of 4 becomes 0 under this rule (4+6=10), and both numbers contain a 4.

40. A) 973 (2→9, 0→7, 6→13→3)

41. 614 (reverse the add-7 rule: 3→6, 8→1, 1→4)

42. 917 (7→9, 9→11→1, 5→7)

43. A) 912 (reverse the add-2 rule: 1→9, 3→1, 4→2)

44. 653 (356 reversed)

45. 184 (481 reversed)

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Homework Worksheet — Answers

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1. BES

2. 235

3. C) SUN

4. RAF

5. 481

6. A) 237

7. TNE

8. 137

9. A) 341, D) 251 and E) 501 are correct (each ends in the digit 1, which codes to T).

10. E) OPT

11. OGL

12. 325

13. B) 256

14. ACT

15. 267

Green — Standard

16. 14 (5 × 2 = 10, 10 + 4 = 14)

17. 8 ((25 − 1) ÷ 3 = 8)

18. D) 19

19. 18 ((4 + 2) × 3 = 18)

20. 16 ((11 − 3) × 2 = 16)

21. C) 7

22. 25 (rule: ×4 then +1; 6 × 4 + 1 = 25)

23. 5 (rule: ×3 then +1; (16 − 1) ÷ 3 = 5)

24. A) 6, B) 7 and C) 8 are correct (codes 15, 17 and 19)

25. A) 33

26. 20 ((5 − 1) × 5 = 20)

27. 7 ((9 + 5) ÷ 2 = 7)

28. B) 17

29. 29 (rule: ×3 then +2; 9 × 3 + 2 = 29)

30. 7 (rule: ×3 then +1; (22 − 1) ÷ 3 = 7)

Standard / Advanced

31. 714 (4→7, 8→11→1, 1→4) — even, since it ends in 4.

32. 839 (2→8, 7→13→3, 3→9) — digit sum 8+3+9 = 20.

33. B) 1 (583 reversed is 385 — only the middle digit, 8, stays in the same place).

34. 862 (reverse the add-4 rule: 2→8, 0→6, 6→2) — 862 is bigger than 206.

35. 819 (918 reversed is 819) — digit sum 8+1+9 = 18.

36. 892 (3→8, 4→9, 7→12→2)

37. 947 (reverse the add-5 rule: 4→9, 9→4, 2→7)

38. A) 137 (731 reversed)

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39. 426 (624 reversed)

40. A) 251 and B) 162 are correct — a digit of 2 becomes 0 under this rule (2+8=10), and both numbers contain a 2.

41. 345 (4→3, 5→4, 6→5, since adding 9 is the same as subtracting 1 with wraparound)

42. 789 (decoding means adding 1 to each digit: 6→7, 7→8, 8→9)

43. A) 509 (905 reversed)

44. 903 (8→9, 9→10→0, 2→3)

45. 609 (reverse the add-1 rule: 7→6, 1→0, 0→9)

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